

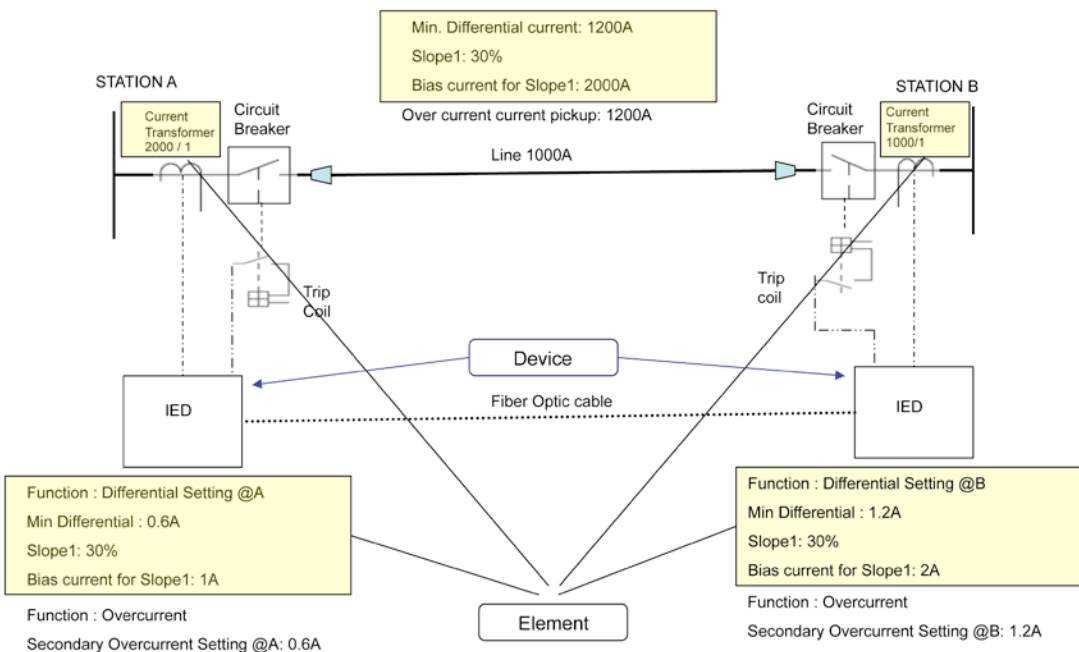
Integrated Platform for Network Element Management

Transforming a traditional grid into a smart grid requires progressive instrumentation through smart devices, as well as the ability to monitor the health of these devices and manage their configuration in a much more holistic way

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As utilities lay out their smart-grid strategies, the inclusion of device management is essential. Since implementing smart grids will require a many-fold increase in the number of devices to be managed, the traditional ways of managing these devices may not scale adequately.

This paper discusses the concept of a network element and its value in terms of device management. It reveals limitations in the way devices are managed today, and illustrates the need for an integrated platform by which to administer them. It further discusses the characteristics of such integrated platforms and the key challenges faced in their establishment.



Caption: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed suscipit ultricies vestibulum. Suspendisse rutrum dictum magna nec lobortis.

What is a Network Element?

An electricity network has several constituents: primary components (also known as power system elements) include generators, transformers, lines and cables, and circuit breakers; secondary components include protection, control and monitoring devices, communication and IT devices.

It is easy to understand what a device is—an Intelligent Electronic Device (IED) shown in the figure below, for example. A set of such devices with specific functionality such as “Minimum Differential” forms a scheme. Such devices can have different settings that can be configured individually. It would have been much easier to manage this configuration or settings across both the devices through one logical entity for this scheme. Called a Network Element, this logical entity is a configuration of a specific function achieved by setting up one or more interlinked devices.

Traditional Grid—How Are Devices Managed Today?

Device management is more of a manual process today. Utility engineers store schemes, device settings and their details either as hard copy or as files during the testing and commissioning phases. Whenever the power network is being modified or there is planned maintenance to configure devices, it is common for engineers to visit the field, implement these changes, and update the files or hard copies.

Device vendors offer proprietary tools to communicate with these devices that work in silos either locally or remotely. These tools usually provide basic functionality (sometimes as elemental as command-line tools) for configuring the device and setting updates when connected.

There are several limitations to the way devices are being managed today:



- **High lead time, effort and cost**

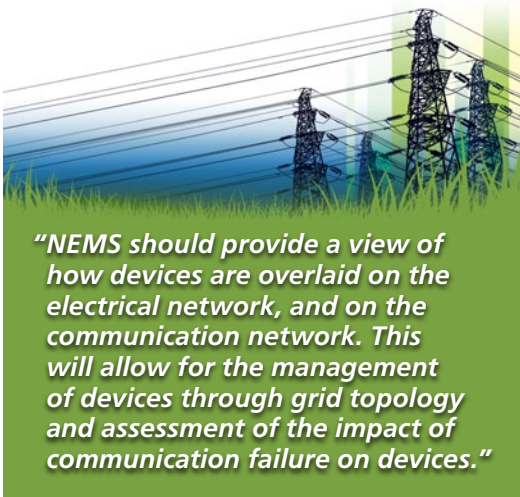
It takes a few weeks for field engineers to carry out maintenance (settings, software or firmware) of devices across the network, especially when this requires making physical visits. There is an associated risk to the network until these changes are implemented. The risk is much higher when the protection coordination for distance, or over current or earth fault relays, needs to be maintained due to dynamic changes in network conditions. With the increase in the number of devices, the cost of manual maintenance and upgrades will also inflate.

- **Lack of governance around maintaining configuration changes**

Configuration changes—firmware versions or settings—are stored manually in files and folders. There is little governance defined around these changes, and the involvement of manual touch points makes the process highly error-prone.

- **Limited visibility into communication network failures**

These devices are interconnected through local- or wide-area communication network components. Hence, limited visibility of communication elements



leads to inefficient and inappropriate results, affecting the overall performance of protection, control and monitoring of devices.

- **Configuration of devices in silos**

Generally, configuration of devices is done through manual calculation that considers various physical inputs such as current, voltage transformer selection, etc. For lack of a facility to manage the interlinked settings through a single logical entity, these devices are configured independently. This may introduce errors and can adversely impact the performance of the power network.

- **Lack of network topology reference**

Product vendor tools do not facilitate the ability to manage devices in the context of grid topology—for example, by substation, by feeder and by interconnection.

Smart Grid—Number of Devices to Manage Will Grow

Consider a 10-MVA, medium-voltage substation consisting of two source feeders, three transformer feeders and ten outgoing circuits. These substations will have approximately 32 measurement devices, 20 intelligent electronic devices for control, protection

and monitoring, telecom routers, and switches and data concentrators at the substation and on the communication network linking to control centers. With the larger addition of distributed energy resources, energy storage systems, static/dynamic Volt-VAR controllers, auto-reclosers and sectionalisers to the distribution network, hundreds of additional devices are expected to be included in each outgoing feeder. It is estimated that, for a typical utility network capacity of more than 1000MW, thousands of devices will need to be managed.

Network Element Management System

It is clear from the above discussion that the traditional ways of managing devices will not scale for a smart-grid context, and this calls for a formal, structured and integrated platform by which to manage devices. This is referred to as a Network Element Management System (NEMS), as defined below.

“NEMS is a platform that allows for defining, managing and monitoring of network elements [refer to the above section ‘What is a Network Element?’], and provides for an integrated view with the power network, communication and asset management system.”

This section captures some of the important characteristics of NEMS.

- **Health Management**

Since the functioning of these elements depends on one or more interconnected devices, it is vital for NEMS to identify and log abnormal instances, and then send appropriate messages to function owners and device owners to take corrective action. NEMS should support the facilities—such as dynamic report by exception to multiple clients, as in IEC 61850 (International Electrotechnical Commission design standard)—in performing effective health management without overloading the communication network.

- **Configuration Management**

The NEMS should have the ability to define and manage elements. It should also have the ability to configure logic for features such as interlock, alarm and trip as per user requirement, to define local alarm or status indicators, and display messages at human-machine interface.

NEMS should further be able to apply the electric network rules automatically while defining and managing elements, thereby maintaining the integrity of the electricity network at all times. NEMS should enable governance around storage of element settings, as well as perform a comprehensive auditing of changes completed, including who made the changes, why and when. It should allow for extraction of current settings, and comparison of existing configuration and settings with revised instructions prior to application to devices. Additionally, it should enhance the ability to study the impact of changing elements on the network.

- **Administration and Analysis**

The NEMS should provide an administrative interface by which to perform user and role-management functions, store security certificates, and update security credentials. It should also facilitate mode selection (e.g., On, Off, Test, Blocked), local or

remote selection, lock-out and tagging of devices. Additionally, functions expected from an NEMS are file management, event and fault analysis, measurement readouts, configurable testing facilities, time synchronizing, etc. The system should also facilitate automatic identification and service provisioning after suitable security verification as the devices are put into service.

- **Performance Management**

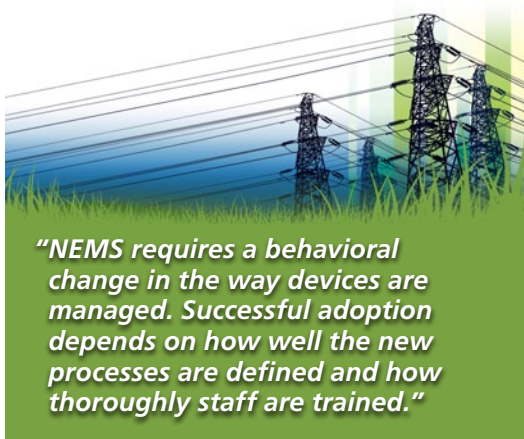
Successful network performance depends on how effectively elements are monitored and managed. NEMS should provide the ability to define Key Performance Indicators (KPI) for networks (e.g., mean time before failure), monitor these devices against the defined KPI, and report the performance against the defined KPI through dashboards.

- **Security Management**

Since the device networks are connected directly through Internet/intranet, securing them from vulnerabilities such as malware, spyware, unauthorized access, etc., is a big challenge. NEMS should have application security, as well as device authentication features, as supported by such standards as IEC 62351.

- **Integrated Visualization**

NEMS should provide a view of how devices are overlaid on the electrical network, and on the communication network. This will allow for the management of devices through grid topology and assessment of the impact of communication failure on devices. NEMS should also provide a view integrated with the asset management system to track the inventory of the devices, with all details (e.g., manufacturer, serial number, date of commissioning, test results, replacement details etc.) accessible so as to maintain the device movement in the network (and even retrieval of the devices).



Challenges

The following challenges should be kept in mind during both conceptualization and the detailed design phase of NEMS.

- **Interoperability and Standardization**

It will be a while before vendors begin supporting the standards maturing in the space of element management. Until then, NEMS will need to deal with underlying vendor legacy, and still be able to provide a consistent view of devices and details.

- **ICT and Embedded Solutions**

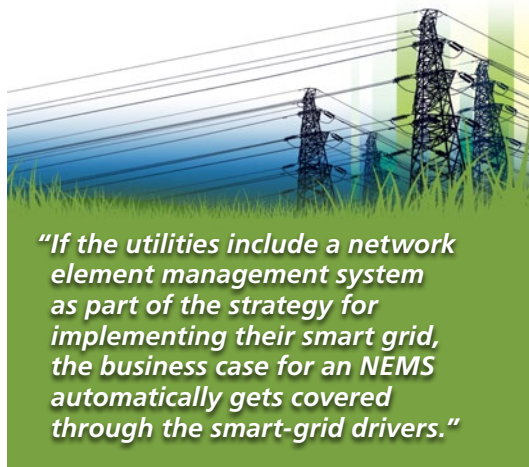
Since the management of network elements involves handshaking between centralized application and network devices, it may not be supported by much of the existing device hardware/firmware. This will call for hardware or firmware upgrades of those devices, necessitating deep involvement by vendors to achieve embedded solutions.

- **Organization Change Management**

NEMS requires a behavioral change in the way devices are managed. Successful adoption depends on how well the new processes are defined and how thoroughly staff are trained.

- **Business Case**

If the utilities include a network element management system as part of the strategy for implementing their smart grid, the business case for an NEMS automatically gets covered through the smart-grid drivers.



Conclusion and Way Forward

Traditional device-management approach will not scale in a smart-grid context, and calls for an integrated platform as discussed through the concept of a Network Element Management System.

This platform should be planned as part of the initial strategy for the smart grid. There are challenges related to interoperability that will be addressed through maturing standards, and a robust architecture that takes legacy devices into account.



To view references and sources please visit www.generatinginsights.com

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